[P4] A seventeen-card witness

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Abstract

An explicit pairwise intersecting 6-uniform family of seventeen cards on 27 symbols with transversal number six. All 136 card pairs intersect, and none of the 80,730 five-symbol subsets is a transversal. Hence $g(6) \leq 17$. This paper uses nothing from [P1], [P2] or [P3]; it can be checked on its own.

This is a witness: it is checked directly and depends on nothing else in the set. No priority is claimed. See the note at the end.

1. The family

Symbols $1, \ldots, 27$.

A_1	1, 2, 3, 4, 5, 21	A_{10}	4, 8, 10, 18, 20, 25
A_2	6, 10, 14, 16, 17, 21	A_{11}	4, 9, 11, 17, 19, 26
A_3	5, 6, 7, 8, 9, 22	A_{12}	5, 12, 15, 16, 20, 26
A_4	1, 13, 15, 17, 18, 22	A_{13}	2, 10, 12, 19, 22, 27
A_5	1, 7, 10, 11, 12, 23	A_{14}	3, 8, 17, 23, 26, 27
A_6	2, 6, 13, 19, 20, 23	A_{15}	4, 9, 15, 16, 23, 27
A_7	2, 8, 11, 14, 15, 24	A_{16}	5, 14, 18, 23, 26, 27
A_8	3, 7, 16, 18, 19, 24	A_{17}	7, 11, 13, 20, 21, 27
A_9	3, 9, 12, 13, 14, 25		

Machine-readable copy: `data/witness_17.json`.

The same family as an incidence matrix, in the format used by Barát and Wanless (arXiv:2011.04444): one row for each symbol $1, \ldots, 27$ in order, one column for each card $A_1, \ldots, A_{17}$ in order, and 1 where the symbol lies in the card. Each column has six ones, and each row's weight is the degree of its symbol.

```
10011000000000000
10000110000010000
10000001100001000
10000000011000100
10100000000100010
01100100000000000
00101001000000001
00100010010001000
00100000101000100
01001000010010000
00001010001000001
```

```
00001000100110000
00010100100000001
01000010100000010
00010010000100100
01000001000100100
01010000001001000
00010001010000010
00000101001010000
00000100010100001
11000000000000001
00110000000010000
00001100000001110
00000011000000000
00000000110000000
00000000001101010
00000000000011111
```

`code/verify_incidence_matrices.py` checks that this matrix is exactly the one built from the data file.

2. The verification

Proposition 1. *The family above is pairwise intersecting, 6-uniform, and has $\tau = 6$. Hence $g(6) \leq 17$.*

Proof. Seventeen distinct cards, each of six distinct symbols, is read off the table. All $\binom{17}{2} = 136$ pairs are checked and every intersection is non-empty. All $\binom{27}{5} = 80{,}730$ five-symbol subsets are checked and none meets all seventeen cards, so $\tau \geq 6$. Since the family is pairwise intersecting, the six symbols of A_1 already meet every card, so $\tau \leq 6$. A cover of size below five would extend to one of size five, so those are excluded too. □

`code/verify_witness_17.py` performs the check in two independent ways — by unions of support bitmasks and by direct set intersection — and needs only the Python standard library. No SAT solver, compiler or external binary is involved.

Do not run the verifier with `python -O`, which disables the assertions it relies on.

3. Structure

The symbol degrees are

$$n_2 = 2, \quad n_3 = 4, \quad n_4 = 19, \quad n_5 = 2.$$

The two degree-five symbols are 23 (in cards 5, 6, 14, 15, 16) and 27 (in cards 13, 14, 15, 16, 17).

The presence of degree-five symbols matters. The sixteen-card search of [P3] imposes a degree-four cap, which is correct there because [P2, Cor 2] forces it at sixteen cards. Imposing the same cap on a seventeen-card search would wrongly discard this family.

The witness is also inclusion-minimal for the property $\tau = 6$: deleting any single card leaves a family with a five-cover. The seventeen deletion covers are stored in `data/witness_17.json` and

are checked by the verifier. That is a property of this particular family; it is not a proof that all sixteen-card families fail, which is what [P3] is for.

4. How it was found

The first twelve cards form a kernel with six degree-two symbols and twenty degree-three symbols, and $\tau = 5$. List all of its five-covers. Introduce a Boolean variable for each, and ask for a selection of at most five of them such that every five-cover of the kernel is disjoint from at least one selected cover — the extension criterion of [P3, Prop 2] with five covers in place of four. Adding a new common symbol, here 27, to each of five selected covers produces five new cards.

The search used Glucose 4 through `python-sat` and returned a selection of five. The verification of the resulting object is independent of that solver: `verify_witness_17.py` re-checks the finished family from scratch.

The search record, with symbols indexed from 0, is kept in `results/search_record.json`. The shift to symbols $1, \ldots, 27$ changes no intersection and no cover number.

5. Verification

claim	script
17 distinct cards, all of size 6	`code/verify_witness_17.py`
all 136 pairs intersect	same
none of the 80,730 five-sets covers	same, by two independent methods
degree histogram	same
inclusion-minimal, with an explicit five-cover per deletion	same

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This work was prepared with AI assistance (ChatGPT, OpenAI; Claude, Anthropic), used for algebraic derivation, for drafting and rewriting code and text, for running and re-running the computations, and for auditing the papers against their own scripts. The same tools were used in the review, so the review carries the same caveat. All statements were checked by the author, who is responsible for them. No priority is claimed and the result has not been independently reviewed by a human or a proof assistant; the repository README sets out the division of labour and the open obligations in full.